

Ground Truth Channel
(only possible in simulation)

4

Causal Inference Output

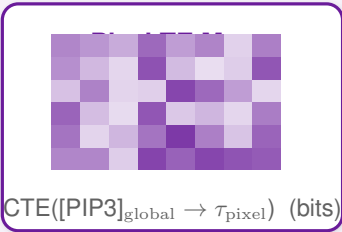
Directed Information Flow Map

First generative simulator bridging molecular dynamics → FLIM-FRET → causal inference

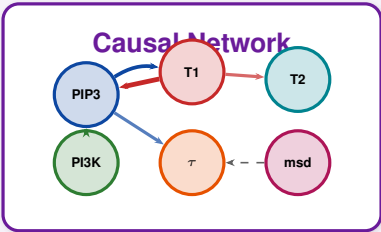
Conditional Causality

$$\text{CTE}(X \rightarrow Y | C)$$

10 orthogonal pairs
Trial-shuffle surrogates
 $\alpha = 0.05$, Wibral criterion



CTE([PIP3]_{global} → τ_{pixel}) (bits)



Causal Network

Time series per replica: $\tau(t)$, $E(t)$, $I(t)$, [PIP3](t), [PTEN](t)

3

FLIM-FRET Image Analysis

7-Layer Quantitative Analysis Stack

L1 Mean τ

L2 Phasor

L3 Bi-exp

L4 STICS

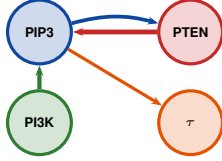
L5 iMSD

★

L6 — Conditional Transfer Entropy (v2)

$$\text{CTE}(X \rightarrow Y | C) = I(Y_{t+1} ; X_t | Y_t, C_t)$$

Resolves confounding by conditioning on observed common drivers



PIP3, PTEN, τ

10 orthogonal pairs

Trial-shuffle surrogates

Wibral lag selection

ADF diagnostic

L7 Error Microscope

$\Delta\tau = |\tau_{\text{image}} - \tau_{\text{exact}}|$ | only possible in simulation

FLIM stack $[n_{\text{ch}} \times n_x \times n_y \times 256]$ per frame

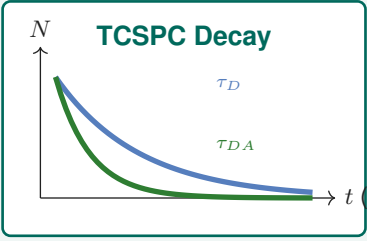
2

Virtual Confocal Microscope

Photon-Level TCSPC Simulation

$$E = \frac{1}{1 + (r/R_0)^6}$$

Förster FRET ($R_0 = 5.74$ nm, mTFP1-Venus)



TCSPC Decay

Poisson shot noise

Photobleaching

PSF + raster scan

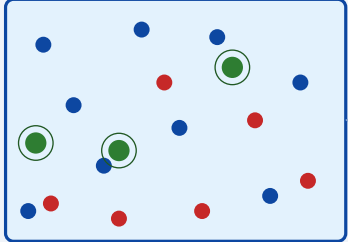
PSS kernel >100×

Positions $\mathbf{r}_i(t)$, species states, bond partners, exact E_i

1

Brownian Dynamics Engine

Stochastic Reaction-Diffusion on 2D Membrane



PIP₃, PTEN, Complex

$$\mathbf{r}(t+dt) = \mathbf{r}(t) + \nabla D dt + \sqrt{2D dt} \boldsymbol{\xi} + \frac{D}{k_{\text{cat}}} \frac{d\mathbf{r}}{dt}$$

Ermak–McCammon (1978) with drift, noise, and force

$$\text{PIP}_3 + \text{PTEN} \rightleftharpoons \text{Complex} \xrightarrow{k_{\text{cat}}} \text{PIP}_2 + \text{PTEN}$$

dt=2 ms

13,000 particles

10×10 μm^2